Experiment -2 Feature Scaling and Data Normalization

Title: Feature Scaling Using Normalization & Standardization

Aim:

To perform scaling of numerical features using:

- Normalization (Min-Max Scaling)
- Standardization (Z-Score Scaling)

Learning Objectives:

After completing this experiment, the student will be able to:

- Understand feature scaling techniques.
- Apply normalization and standardization.

Software Requirements:

Python 3.x, Jupyter Notebook / Anaconda / Colab

Required libraries:

pandas, numpy, scikit-learn

if you are using pip	→ pip install scikit-learn
if you are using conda	→ conda install scikit-learn
if you are using jupyter notebook	→ !pip install scikit-learn

Theory:

Different features may have different ranges. To bring all the features in to the same range, we can use normalization or standardization techniques. There are many methods for data normalization.

- **min-max normalization:**
  - performs a linear transformation on the original data.
  - Suppose that min_A and max_A are the minimum and maximum values of an attribute A.
  - Min-max normalization maps a value x_i of the attribute A to $x_{i,n}$

$$x_i' = \frac{x_i - min_A}{max_A - min_A}(new_max_A - new_min_A) + new_min_A$$

- ***z*-score normalization/ standardization** (or *zero-mean normalization*)**:**
  - the values for an attribute *A*, are normalized based on the mean (i.e., average) and standard deviation of *A*.
  - z-score normalization maps a value x_i of the attribute A to x_i'

$$z_i = \frac{x_i - \bar{A}}{\sigma_A}$$

where $\bar{A}$ and σ_A are the mean and standard deviation, respectively, of attribute *A*.

Note: Expand the theory part based on the class room knowledge.

Functions Used

Function	Purpose
df.filling()	Fills missing values
df.drop_duplicates()	Removes duplicate rows
MinMaxScalers()	MinMax normalization
StandardScaler**()**	Z-score normalization
minmax.fit_transform()	Fits the scaler to the data
df.copy()	Creates a copy of the DataFrame so that changes don't affect the original data.
head()	Displays the first 5 rows of the DataFrame
tail()	Displays the last 5 rows of the DataFrame

Procedure/ Algorithm:

1. Import libraries
2. Load dataset/ generate data
3. Apply normalization
4. Apply standardization
5. Visualize the normalized data

Program:

Output:

Conclusion:

In this experiment, applied normalization and standardization techniques to the given dataset.

Sampled Viva Questions:

1) What is feature scaling?
2) Difference between normalization and standardization?
3) Why is scaling important in ML?
4) What is MinMaxScaler?
5) What is StandardScaler?
6) What is a data pipeline?